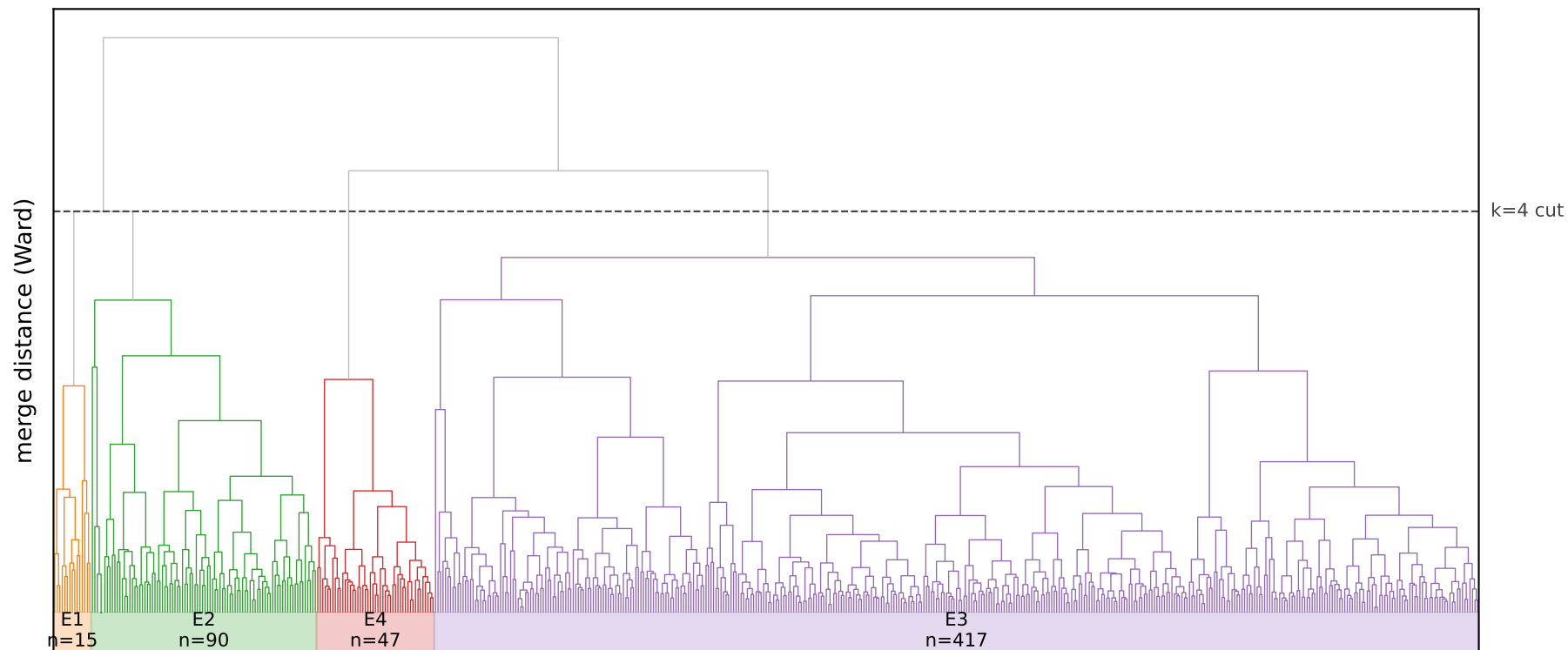


EXAM behavior: Ward dendrogram, coloured at the k=4 cut (n=569)

each cluster is described below by its feature signature; the up/down value is how far its students sit from the whole-group average, in standard deviations



E1 (n=15)

- ▼ distinct concepts -3.4
- ▲ edits, no score gain +2.9
- ▼ attempts per question -2.2
- ▲ lines changed / edit +1.6
- exam grade mean 1 · med 0

E2 (n=90)

- ▼ code lines -1.4
- ▼ distinct concepts -1.0
- ▼ tries that improved -1.0
- ▼ lines changed / edit -0.5
- exam grade mean 12 · med 4

E4 (n=47)

- ▲ small fix, big gain +2.8
- ▲ tries that improved +0.8
- ▲ questions attempted +0.6
- ▲ code lines +0.5
- exam grade mean 85 · med 90

E3 (n=417)

- ▲ code lines +0.3
- ▲ distinct concepts +0.3
- ▼ small fix, big gain -0.2
- ▲ tries that improved +0.2
- exam grade mean 40 · med 33

Clusters are formed from the behavior features only; the exam grade (mean final score over the 6 exam questions, 0 to 100) is shown only to relate the groups to the outcome.